

Explainable SHAP-XGBoost models for Mental Health in the COVID-19 Lockdown Period In Italy

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Abstract. The COVID-19 pandemic caused the worst health crisis in history. Many countries implemented restrictive policies to control the disease through social distancing. While this policy proved effective in controlling the disease, social distancing also had negative consequences for mental health. Numerous papers have utilized machine learning to predict and explain outcomes, highlighting the impact of specific expected results. However, the data collected comes from different research groups and regions, leading to a lack of globally consistent data. Therefore, it is difficult to see factors common to all populations worldwide. This study, based on 1166 observations of the Italian population, aims to provide a more comprehensive understanding of the factors influencing expected machine learning outcomes across various datasets. This research focuses on solving multivariate regression problems using XGBoost and analyzing the factors that affect prediction results using SHAP (Shapley Additive exPlanations) analysis. Experimental results show that the model achieved high accuracy with a low mean absolute error (MAE) of approximately 1 for the response variables. In particular, the study indicated that private space at home and isolation time were the two variables with the most decisive influence on mental health. This finding advances the technique of predicting mental health and demonstrates how attributes affect prediction outcomes using the XGBoost machine learning model. The study's findings contribute to our knowledge of how social factors affect people's mental health when they are socially isolated.

Keywords: Mental health · COVID-19 · Machine learning · Explainable Artificial Intelligence · Explainable SHAP-XGBoost

1 Introduction

In public health emergencies, social distancing and isolation measures are often implemented to limit the spread of disease. However, in addition to their effectiveness in controlling the epidemic, these measures also have significant

negative impacts on people’s mental health. Numerous studies have shown that limited social contact, changes in daily routines, and increased economic instability can increase levels of stress, anxiety, and depression in the community [1–4]. Therefore, analyzing the factors affecting mental health in the context of social isolation is a matter of significant scientific and practical importance.

Recently, numerous studies have utilized survey data to assess the impact of social distancing during the COVID-19 pandemic on people’s mental health. Most of these studies focused on analyzing the relationship between socioeconomic factors, demographics, and living conditions with mental health indicators using traditional statistical methods such as linear regression [5], logistic regression [6, 7], or structural equation modeling [8, 9]. The results obtained have contributed to clarifying the role of factors such as age, gender, employment status, level of social support, and housing conditions in mental health during the isolation period.

However, traditional regression methods often rely on linear assumptions and are limited in capturing nonlinear relationships or complex interactions between multiple explanatory variables. In the context of mental health data being simultaneously influenced by many different factors, these assumptions can lead to an underestimation of the true role of each characteristic. While machine learning models have the potential to improve predictive accuracy and handle nonlinear relationships effectively, their application in mental health research remains relatively limited, primarily due to difficulties in interpreting the results.

In recent years, Explainable Artificial Intelligence (XAI) has been proposed as an approach to overcome this limitation [10]. XAI techniques enable the explanation of the contribution of each input variable to the predictive results of a machine learning model, thereby enhancing transparency and interpretability. In particular, SHapley Additive exPlanations (SHAP) has been introduced as a framework for explaining complex model predictions by quantifying the contribution of each feature to the outcome [11]. Currently, many studies combine powerful machine learning models, such as Extreme Gradient Boosting (XGBoost), with XAI methods like SHAP and LIME to analyze and explain factors influencing mental health in the context of large-scale social isolation [12, 13]. However, these studies are based on separate surveys from different research groups in specific localities or territories. Therefore, the data are sometimes inconsistent, leading to potentially different conclusions across different datasets. This raises the question of whether there are any characteristics or traits common to many different territories.

Over the past few years, numerous research groups have applied machine learning models combined with explainable artificial intelligence techniques to predict mental health outcomes and explain influencing factors during the COVID-19 pandemic. For example, XGBoost regression combined with SHAP analysis was used to identify key predictors of emotional stress, such as financial anxiety and fear of infection, across multiple survey periods [14]. Similarly, poor mental health status in adult Indians during the pandemic was examined using XGBoost and SHAP to reveal leading risk factors such as sleep disturbance, age,

general health, income, wearing a mask, and self-reported discrimination [13]. Interpretable machine learning models have also been successfully applied in predicting depression, showing that variables such as anxiety and lifestyle factors contribute significantly to the model’s prediction [12]. Additionally, this method has been applied to assess mood changes in children and adolescents during lockdowns [15]. Interpretable models combined with SHAP have been used to classify and identify risk factors for anxiety, depression, and insomnia, demonstrating the broad applicability of these methods in mental health research following the COVID-19 lockdown [16].

The predictive power of machine learning in mental health research is well-established, yet it is the inclusion of cross-regional datasets that uncovers the specific socio-environmental factors at work. Having undergone an exceptionally strict and lengthy period of lockdown, Italy provides a case study for examining the societal psychological toll. This study contributes by combining the non-linear predictive model XGBoost with SHAP interpretive analysis to clarify the relative and interactive roles of living conditions, demographic and social characteristics, and epidemiological context indicators. This study not only focuses on the accuracy of prediction but also demonstrates the non-linear relationship between the predictor variable and the outcome variable. It provides policymakers with the knowledge to adjust their policies when future health crises occur.

2 Materials and Methods

2.1 Data Preprocessing

The data used in this study were collected from an online psychological survey conducted by the Department of Psychology at the University of MilanoBicocca (Italy) during the first lockdown phase of the COVID-19 pandemic.¹ The original dataset was described and analyzed in the study by Bonnini and Borghesi (2022). [17].

Initially, the survey recorded 2,470 accesses. After retaining only participants who confirmed informed consent to participate in the study (*InformedConsent* = 1) and excluding cases in which the questionnaire was not completed, the dataset was reduced to 2,120 observations. Records containing missing values in the original variables were subsequently removed using a listwise deletion approach, resulting in 2,107 observations.

The data were then standardized in terms of formatting, and analytical variables were constructed. All quantitative variables and scale-based items were converted into numeric form to facilitate subsequent analyses. Four mental health outcome variables and fifteen predictor variables were selected following the original study that employed the same dataset, in order to preserve conceptual comparability while allowing the present research to focus on methodological

¹ The survey dataset is published on OSF: https://osf.io/xb8yj/?view_only=966abafccc844b99924da85be3f76272

differences in the analytical approach. The predictor variables include indicators of living space conditions, demographic characteristics, social factors, and measures reflecting the epidemiological context.

During the construction of analytical variables specifically, the computation of scale means and epidemiological ratios additional observations with missing values emerged and were excluded, resulting in a final sample size of $N = 1314$. No data imputation procedures were applied, in order to maintain consistency with the reference study and to avoid introducing additional, potentially unwarranted assumptions into the data. The four mental health outcome variables *Depression*, *Unworthiness*, *Alienation*, and *Helplessness* were constructed by computing the mean score of five Likert-scale items, following the same approach adopted in the original study. To ensure the reliability of the measurement, mean scores were calculated only for participants who provided valid responses to all five items; observations with any missing item were excluded from the analytical dataset.

The mental health outcome variables (*Depression*, *Unworthiness*, *Alienation*, and *Helplessness*) were computed as the mean scores of five Likert-scale items and were applied only for participants with complete responses to all items. For the epidemiological variables, inconsistencies in data formatting were identified during data verification. Consequently, exposure indicators were recalculated and expressed as relative ratios, including $I_{\text{Prov/Reg}}$ (the ratio of provincial-level cases to regional-level cases), $I_{\text{Reg/Country}}$ (the ratio of regional-level cases to national-level cases), and $I_{\text{Prov/Country}}$ (the ratio of provincial-level cases to national-level cases).

To support model training, we further required complete data across all predictor and outcome variables used in the XGBoost analysis. This step resulted in a final sample size of $N = 1166$. The analytical dataset comprised four outcome variables and fifteen predictor variables, for a total of 19 variables. The predictor variables were grouped into four main categories: (i) living space conditions (*SpaceAdequacy_Size*, *SpaceAdequacy_Brightness*, *SpaceAdequacy_Privacy*); (ii) demographic characteristics (*Gender*, *Age*, *Occup*, *Educ*); (iii) social factors (*SocialIsolation*, *NumberOfContacts*, *ProportionOfflineContacts*, *OfflineContacts_Closeness*, *OnlineContacts_Closeness*); and (iv) epidemiological context indicators $I_{\text{Prov/Reg}}$, $I_{\text{Reg/Country}}$, and $I_{\text{Prov/Country}}$.

This dataset was used directly for training and evaluating the XGBoost models. As the underlying model is tree-based, no feature scaling was applied to the predictor variables. In addition, no separate outlier treatment was applied, as tree-based methods are well suited to handling skewed distributions and extreme values commonly observed in survey data.

Isolation Duration During the exploratory data analysis stage, we proceeded to examine the descriptive relationship between the duration of social isolation and the mental health indicators. The visualized results indicate that isolation duration is closely associated with variations in the psychological outcomes considered. Overall, the levels of *Depression*, *Unworthiness*, *Alienation*, and *Helplessness* tend to increase as the length of isolation extends, with particularly

pronounced changes observed during the period approximately between 5 and 15 days.

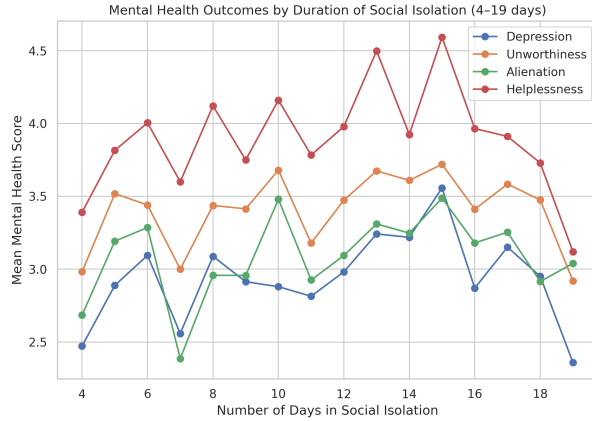


Fig. 1: Descriptive trends of mental health outcomes across increasing durations of social isolation (4-19 days).

Among these aspects, *Helplessness* exhibits both the highest overall level and the most pronounced increase, reaching a peak around days 1315. This pattern suggests that feelings of helplessness may represent a particularly salient psychological response in the context of increasing isolation. The remaining aspects also reach their highest values within the same period, although with lower levels of variation. After the mark of approximately 1516 days, the observed indicators tend to show a slight downward trend. This pattern may reflect the emergence of psychological adaptation processes over time, or it may be related to differences in sample composition at longer isolation durations. However, these interpretations are exploratory in nature and warrant careful consideration in subsequent quantitative analyses.

2.2 Methods

XGBoost regression model In this study, we employ Extreme Gradient Boosting (XGBoost) [18] regression models to predict each mental health outcome variable. XGBoost is an ensemble learning algorithm based on decision trees, in which trees are constructed sequentially through a gradient boosting mechanism to progressively improve predictive accuracy. This approach is capable of capturing nonlinear relationships and complex interactions among predictor variables without requiring prior assumptions about the functional form linking independent and dependent variables. Such properties make XGBoost particularly well suited for social survey data, where relationships between factors are often nonlinear and difficult to adequately capture using traditional

parametric models. Within the XGBoost framework, the predicted value for observation i is expressed as the sum of the outputs from M regression trees:

$$\hat{y}_i = \sum_{m=1}^M f_m(\mathbf{x}_i), \quad f_m \in \mathcal{F}, \quad (1)$$

where $\mathbf{x}_i$ denotes the predictor vector for observation i , and $\mathcal{F}$ represents the space of decision trees.

Objective function The model training process was carried out by minimizing a regularized objective function, which consists of a loss term and a model complexity control term:

$$\mathcal{L} = \sum_{i=1}^n \ell(y_i, \hat{y}_i) + \sum_{m=1}^M \Omega(f_m), \quad (2)$$

where $\ell(\cdot)$ denotes the loss function, specified as the squared error loss in this study, and $\Omega(\cdot)$ represents the regularization term used to limit the complexity of individual trees.

Hyperparameter optimization using Optuna Hyperparameter optimization was conducted using the Optuna library [19], a modern Bayesian optimization framework. Optuna employs the Tree-structured Parzen Estimator (TPE) sampler to efficiently explore the hyperparameter space, allocating greater search effort to regions associated with higher model performance compared to conventional grid search or random search approaches.

A separate optimization procedure was conducted for each outcome variable, with 500 trials performed for each model. In each trial, the hyperparameter configuration was evaluated using a 5-fold cross-validation strategy, in which the dataset was randomly partitioned into five folds, with four folds used for model training and one fold reserved for validation. The mean absolute error (MAE), averaged across the five folds, was used as the objective function to be minimized. A fixed seed was employed to ensure the reproducibility of the optimization process.

Training procedure After identifying the optimal hyperparameter configuration for each outcome variable, the final XGBoost model was retrained on the full dataset. This approach allows for maximal utilization of the available observational information, while ensuring that model configurations were selected based on out-of-sample evaluation through cross-validation during the optimization stage.

Model Interpretation Using SHAP Although XGBoost achieves strong predictive performance, the nonlinear and ensemble-based structure of the model limits the direct interpretability of the relationships between predictor variables and outcomes. Therefore, to enhance post hoc interpretability following the prediction stage, we apply the SHAP [11] framework as a model-agnostic explanation layer within the explainable artificial intelligence (XAI) paradigm.

SHAP decomposes the model prediction into the sum of individual contributions from each predictor variable, based on Shapley values derived from cooperative game theory. For a given observation i , the model prediction can be expressed as:

$$\hat{y}_i = \phi_0 + \sum_{j=1}^p \phi_{ij}, \quad (3)$$

where ϕ_0 denotes the mean model prediction over the entire sample, and ϕ_{ij} represents the SHAP value of predictor j for observation i . SHAP enables model interpretation at two complementary levels. At the global level, variable importance is assessed through the mean absolute SHAP values computed across the entire sample, reflecting the overall influence of each predictor on the model predictions. At the local level, SHAP quantifies the contribution of each variable to individual observations, while also indicating the direction of the effect that is, whether a given predictor increases or decreases the predicted outcome.

3 Result

3.1 XGBoost Models

In this section, we report the predictive performance of the XGBoost models for the four mental health outcome variables, evaluated using the MAE and RMSE metrics. The results presented in Table 1 indicate that the XGBoost models

Table 1: Predictive performance of the XGBoost models and comparison with the adjusted R^2 values reported for linear regression models in the original study.

Mental Health	MAE	RMSE
Depression	1.202	1.439
Unworthiness	0.924	1.13
Alienation)	1.136	1.372
Helplessness	1.195	1.466

achieve relatively consistent predictive performance across all four mental health outcomes. Specifically, MAE values range from 0.801 to 1.026, corresponding to approximately 13.2 – 17% of the full 1 – 7 scale. This suggests that, on average, prediction errors are close to one scale point. RMSE values fall within the range of 1.13 to 1.466, reflecting a relatively stable level of overall prediction error across the different outcomes.

When examined by outcome, the model predicting *Unworthiness* achieved the lowest MAE (0.924) and RMSE (1.13), indicating relatively higher predictive accuracy compared to the other mental health outcomes. In contrast, *Helplessness* exhibited the highest prediction error, with an MAE of 1.195 and an RMSE of 1.466, suggesting greater difficulty in prediction in terms of both average absolute error and squared error. The remaining outcomes, *Depression* and *Alienation*, showed intermediate levels of prediction error, with relatively similar MAE and RMSE values. Overall, the predictive performance of the models is relatively consistent across the mental health outcomes, with no substantial disparities in error magnitude. The differences identified across outcome dimensions suggest that predictive accuracy may vary depending on the specific characteristics of each psychological aspect. At the same time, these differences provide a basis for more detailed interpretative analyses of the predictive structures in the subsequent sections.

Distribution of prediction errors Figure 2 presents the distribution of prediction errors produced by the XGBoost models for the four mental health outcomes, where the error is defined as the difference between the predicted and observed values ($\hat{y} - y$). Overall, the median errors for all four outcomes are



Fig. 2: Distribution of prediction errors for the XGBoost models across the four mental health outcomes ($\hat{y} - y$).

close to zero, indicating the absence of pronounced systematic bias in the model predictions.

However, the error distributions reveal the presence of a number of observations with relatively large prediction deviations, predominantly on the negative error side, indicating cases in which the model substantially underestimates the observed values. This pattern is more pronounced for the *Depression* and *Alienation* outcomes, whereas the error distributions for *Unworthiness* and *Helplessness* are more tightly concentrated around the central value.

In terms of overall dispersion, *Unworthiness* exhibits a narrower interquartile range, indicating more stable typical prediction errors, whereas *Depression* and *Alienation* exhibit longer error tails, indicating that prediction errors are not evenly distributed across individuals. These distributional characteristics are consistent with the MAE and RMSE values reported earlier, while also providing a more detailed view of the stability and variability of prediction errors at the individual-observation level.

3.2 SHAP Value

SHAP results for Depression The SHAP analysis shows that predictions of Depression are mainly driven by living conditions and social connectedness. The most influential variables are *SpaceAdequacy_Privacy*, *SpaceAdequacy_Brightness*, *SocialIsolation*, *NumberOfContacts* and *SpaceAdequacy_Size*. Among these, *SpaceAdequacy_Privacy*.

Lower levels of residential privacy, poorer lighting conditions, smaller living space, and higher levels of social isolation are all associated with positive SHAP values, corresponding to higher predicted levels of depression. In contrast, better housing conditions and broader social networks, reflected by a larger number of social contacts, are associated with negative SHAP values, indicating a protective effect. Across all variables, the majority of observations are concentrated around SHAP values close to zero. This pattern suggests that these influences are widespread across the sample but are generally of moderate magnitude, rather than being driven by extreme effects.

Overall, the SHAP patterns suggest that depressive symptoms are mainly associated with concurrent constraints in both the physical and social environments. Housing inadequacy and social isolation appear to operate in parallel, with each factor contributing incrementally to higher predicted depression. This highlights the combined role of environmental comfort and social relational resources in shaping psychological vulnerability.

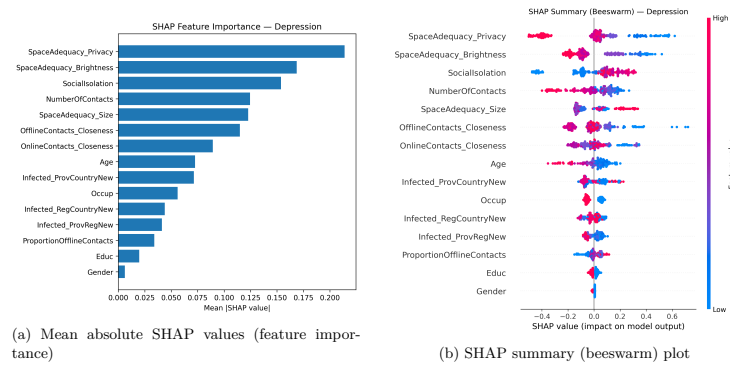


Fig. 3: SHAP-based interpretation of the XGBoost model for Depression.

SHAP Results for Unworthiness For Unworthiness, SHAP values indicate that the model is primarily influenced by *Age*, *SpaceAdequacy_Privacy*, *OfflineContacts_Closeness*, *SpaceAdequacy_Brightness* and *NumberOfContacts*.

Younger age, lower levels of residential privacy, weaker closeness in offline social relationships, and poorer lighting conditions are all associated with positive SHAP values, contributing to higher predicted levels of Unworthiness. In contrast, older age, better housing conditions, and closer social relationships correspond to negative SHAP values, indicating a protective effect. Similar to the results observed for Depression, SHAP values for Unworthiness are largely concentrated around zero. This pattern suggests that these effects are broadly distributed across the sample but are generally of moderate magnitude. The variable *NumberOfContacts* also exhibits an inverse relationship with the predicted outcome, whereby a larger number of social contacts is associated with negative SHAP values, while fewer contacts correspond to positive SHAP values. Most observations are clustered within the SHAP range from -0.1 to 0.1 , indicating that this effect is common across the sample.

Overall, the most influential factors for Unworthiness reflect a combination of age-related characteristics, living conditions, and the quality of social connectedness. Less favorable living environments, weaker social ties, and younger age tend to be associated with higher predicted Unworthiness, highlighting the joint contribution of individual developmental factors and environmental context to feelings of low self-worth.

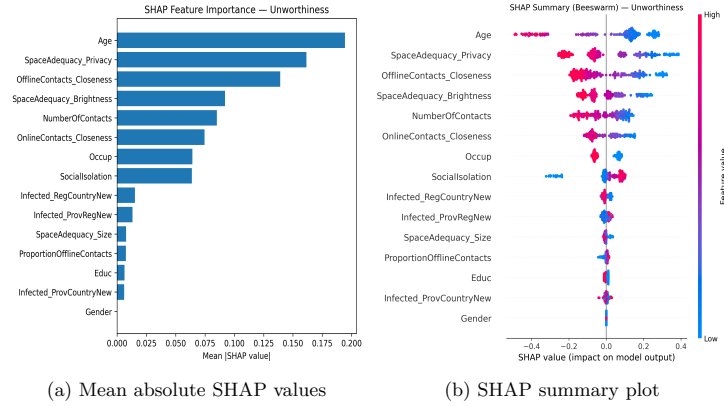


Fig. 4: SHAP interpretation for Unworthiness

SHAP results for Alienation The most important predictors for Alienation include *Age*, *OnlineContacts_Closeness*, *SpaceAdequacy_Privacy*, *NumberOfContacts* and *OfflineContacts_Closeness*. Overall, the SHAP patterns indicate that feelings of alienation are shaped by the combined effects of life stage, housing conditions, and the perceived quality of both online and offline social

relationships. Younger individuals appear to be more psychologically vulnerable to experiences of social separation, whereas emotionally close relationships particularly in online contexts play a substantial protective role.

Residential privacy continues to function as an environmental resource, suggesting that limitations in personal living space may increase perceptions of social disconnection. Taken together, these results emphasize that Alienation is not dominated by a single domain, but rather reflects the interaction between individual developmental factors and the structure of social relationships in both digital and face-to-face settings.

Although the magnitude of SHAP effects is generally moderate, their consistency across a large proportion of the sample indicates that these variables represent stable and common contributors to the models predictions.

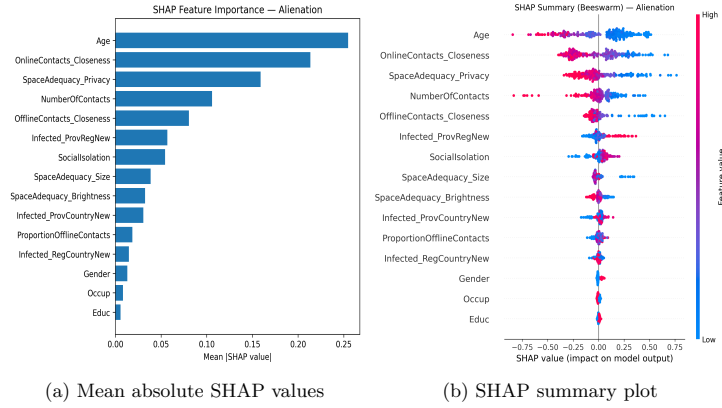


Fig. 5: SHAP interpretation for Alienation

SHAP results for Helplessness For Helplessness, SHAP values indicate that the predictive structure of the model is primarily influenced by *SpaceAdequacy_Privacy*, *OnlineContacts_Closeness*, *SpaceAdequacy_Brightness* and *OfflineContacts_Closeness*.

Higher levels of social isolation, lower residential privacy, weaker emotional closeness in both online and offline relationships, and poorer lighting conditions are all associated with positive SHAP values, contributing to higher predicted levels of Helplessness. In contrast, lower social isolation, better housing conditions, and closer social relationships correspond to negative SHAP values, indicating protective effects. Similar to the other outcomes, SHAP values are predominantly concentrated around zero, suggesting that these influences are widely distributed across the sample but generally of moderate magnitude.

Overall, the most important factors associated with Helplessness reflect the combined roles of social disconnection and living environment conditions. Limited social support networks and constrained housing environments tend to be

linked to higher predicted levels of Helplessness, underscoring the joint importance of interpersonal resources and residential context in shaping perceptions of loss of control.

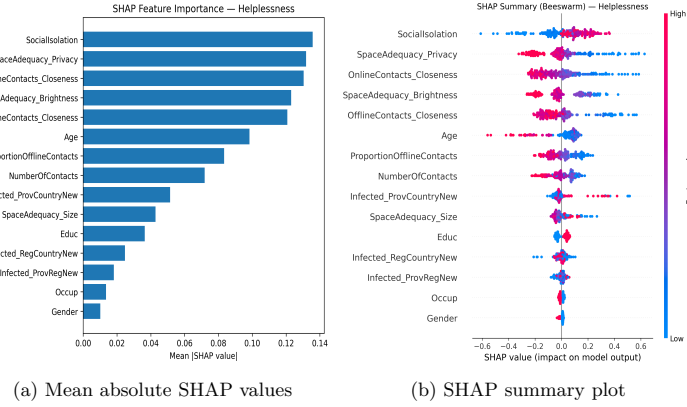


Fig. 6: SHAP interpretation for Helplessness

4 Conclusion

The findings of this study indicate that mental health during lockdown periods is closely associated with living environment conditions and levels of social connectedness. Factors such as lighting conditions, residential privacy, and the quality of social relationships show consistent associations across multiple dimensions of mental health, suggesting that psychological experiences during the pandemic are shaped not only by individual characteristics but also by the surrounding living context. From a policy perspective, these results highlight the necessity of integrating mental health protection into the design of public health interventions in emergency situations.

This study has several important limitations that should be considered when interpreting the results. First, the cross-sectional nature of the data, together with the reliance on self-reported measures and online voluntary sampling, limits the ability to infer causal relationships and reduces the overall generalizability of the findings. Second, although machine learning models enable the identification of complex associations, the predictive performance remains relatively modest, reflecting the multidimensional and inherently difficult-to-capture nature of mental health outcomes.

Future research employing longitudinal data and explicitly incorporating socio-cultural contexts would allow for a clearer understanding of how mental health evolves over time. Moreover, integrating evidence on mental health into the preparation and design of lockdown policies across different national contexts may help mitigate adverse psychological consequences during public health emergencies.

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